ModelIt!

Student Activity Pack

Can Nanotechnology Deliver Medicine to Individual Cells?

Modeling Targeted Drug Delivery, Nanoparticle Design, and Therapeutic Precision

Grade 11L3 | NGSS HS-LS1-3, HS-PS2-4 | Scale, Proportion, and Quantity

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you already know about how cancer is treated with chemotherapy, and why does chemotherapy cause side effects in healthy parts of the body?**

2. **If you could design a medicine that only affected cancer cells and left healthy cells alone, what properties would it need to have?**

3. **Draw a diagram showing how you think a drug-carrying nanoparticle could find and deliver medicine to a specific cancer cell in the body.**

4. **What is the difference between a drug that spreads evenly throughout the body versus one that concentrates at the site of disease?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Nanoparticle Size | Surface PEGylation | Targeting Ligand Density | Immune Clearance Rate | Tumor Accumulation | Drug Release Rate | Off-Target Toxicity

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: PEGylated Passive Targeting

100nm nanoparticle, dense PEG coating, no targeting ligands, EPR-dependent accumulation

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Active Targeting with Antibody Ligands

100nm nanoparticle with PEG coating and tumor-specific antibody ligands

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Stimulus-Responsive Release

pH-sensitive nanoparticles stable at blood pH 7.4 but releasing drug at tumor pH 6.5

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Blood-Brain Barrier Penetration — The ability of nanoparticles to cross the highly selective blood-brain barrier — a layer of tightly joined endothelial cells that prevents most molecules from entering brain tissue, requiring specialized surface modifications like transferrin receptor-targeting ligands or temporary barrier disruption techniques

Combination Payload — Nanoparticles carrying multiple therapeutic agents — such as a chemotherapy drug plus an immune-stimulating molecule — that can be released at different rates to achieve synergistic effects at the tumor site that neither drug achieves alone

Theranostic Capability — Nanoparticles that combine therapeutic drug delivery with diagnostic imaging — containing both a drug payload and an imaging contrast agent so that doctors can visualize exactly where the particles accumulate and track treatment response in real time

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Can we design nanoparticles that navigate the human body, find specific diseased cells, and deliver medication directly to them — turning the blunt instrument of chemotherapy into a precision tool? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate why the 0.7% tumor delivery rate is both disappointing compared to the vision and impressive compared to conventional chemotherapy?**

2. **Why does the PEGylation dilemma — needing stealth for survival but losing stealth for targeting — represent a fundamental design trade-off that cannot be fully resolved?**

3. **What does your model reveal about why active targeting improves cellular uptake specificity but does not dramatically increase the total amount of drug reaching the tumor?**

4. **How would your model change if nanoparticles could be designed that are completely invisible to the immune system?**

5. **What are the limitations of modeling nanomedicine biodistribution when individual patients' immune systems, tumor vasculature, and blood chemistry vary significantly?**

8 STEM Challenge: Design a Nanoparticle Drug Delivery System

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YOUR ENGINEERING MISSION

Design a nanoparticle drug delivery system for a specific cancer type, addressing particle design, targeting strategy, release mechanism, and predicted therapeutic advantage over conventional chemotherapy.

THE SCENARIO

An oncology research lab has challenged your team to design a nanoparticle drug delivery system for one of three cancer types: brain cancer (requires crossing the blood-brain barrier), pancreatic cancer (dense stromal tissue that blocks nanoparticle penetration), or melanoma (accessible tumors but high metastatic spread). Each cancer presents unique delivery challenges that your nanoparticle design must address. You must specify particle size, surface chemistry, targeting ligand, drug payload, release mechanism, and predicted biodistribution — and honestly assess how much improvement your design offers over conventional treatment.

Engineering Design Thinking

1. WHAT MATERIAL WOULD YOU CHOOSE FOR THE NANOPARTICLE CORE — LIPID, POLYMER, SILICA, OR METAL — AND WHAT ARE THE BIOCOMPATIBILITY AND DRUG LOADING TRADE-OFFS?

2. WHAT TARGETING LIGAND WOULD YOU SELECT FOR YOUR SPECIFIC CANCER TYPE AND WHAT RECEPTOR DOES IT BIND ON THE CANCER CELL SURFACE?

3. HOW WOULD YOU ADDRESS THE PEGYLATION DILEMMA — THE NEED FOR STEALTH COATING VERSUS THE NEED FOR TARGET CELL UPTAKE?

4. WHAT PRECLINICAL TESTS WOULD YOU PROPOSE TO EVALUATE YOUR NANOPARTICLE SYSTEM'S SAFETY AND EFFICACY BEFORE HUMAN TRIALS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Nanomedicine Scientists design and characterize nanoparticle drug delivery systems at pharmaceutical companies and research institutions, earning $90,000-$170,000/year. Pharmaceutical Engineers scale up nanoparticle manufacturing from laboratory batches to clinical production volumes while maintaining quality control, earning $85,000-$160,000/year. Clinical Pharmacologists evaluate how nanomedicines behave in human patients — tracking biodistribution, efficacy, and side effects — to optimize dosing and treatment protocols, earning $100,000-$200,000/year.